

PROJECT-BASED LEARNING

PHASE-I EVALUATION

**SlopeShield-Intelligent Land-
slide Risk & Safe Route System**

Team ID: DSCPP- III- 2026 -T099

Department of Computer Science & Engineering

Graphic Era (Deemed to be University), Dehradun

Academic Session 2026-27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T099

Semester: 3rd

Section: ML-2,ML-3,ML-4,DS-2

Domain: Disaster Management/Smart T

Team Members

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Mentor

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Interactions completed:4

Problem Statement & Motivation

Problem Statement

Hilly regions are highly vulnerable to landslides, especially during the heavy rainfall. Landslides can block roads, damage infrastructure, and endanger travellers and emergency services.

SlopeShield identifies landslide-prone areas using rainfall, slope, soil moisture, elevation, historical landslide records and road conditions. It also recommends safer routes by considering the risk of different road segments.

Motivation

- ▶ Heavy rainfall can increase landslide risk.
- ▶ Landslides can block roads.
- ▶ Navigation mainly consider distance and traffic.
- ▶ Travellers need safer route options.

Target Users / Stakeholders

- ▶ Travellers in hilly regions
- ▶ Emergency response teams
- ▶ Road and disaster management authorities
- ▶ Users requiring safer route recommendations

Objectives

Project Objectives

1. Collect and organize environmental and road-related data.
2. Develop a landslide risk scoring system using rainfall, slope, soil moisture, elevation, historical landslide records and road conditions.
3. Classify areas into Low, Moderate, High and Critical risk levels.
4. Represent the road network as a weighted graph and use Dijkstra's algorithm with a min-heap priority queue to recommend safer routes.
5. Develop a web interface to display landslide risk zones and recommended routes.

Key Objective: Combine landslide risk assessment with route recommendation to help users choose safer routes in hilly regions.

Proposed Solution

Proposed Approach

1. *Input / Data Collection*
Weather, slope and historical data.
2. *Processing / Core Algorithm*
Calculate and classify landslide risk.
3. *System Logic*
Represent roads as a weighted graph.
4. *Database / Storage*
Dijkstra with priority queue.
5. *Output / Visualization*
Show risk zones and routes.

Key Features

- ▶ Landslide risk scoring
- ▶ Four-level risk classification
- ▶ Weighted road network graph
- ▶ Dijkstra-based safe route recommendation
- ▶ Interactive map using Leaflet and OpenStreetMap

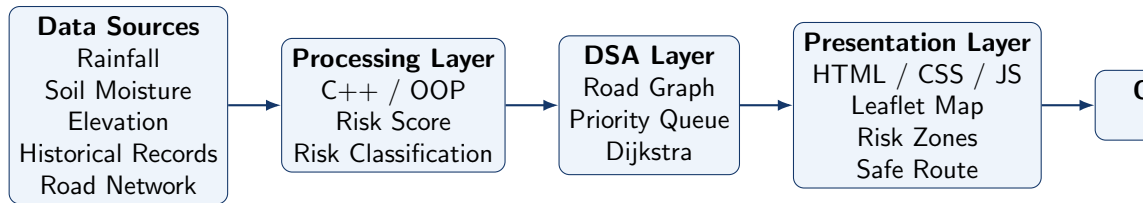
The system combines risk assessment with route planning.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	Graphs, vectors, priority queue / min-heap, hash table	Road Network & Route Engine
OOPs with C++	Classes, objects, encapsulation, Constructors and STL	Risk, Road & Route Modules

Requirement: DSA concepts are used for graph-based route processing, while C++ OOP is used to model locations, roads, risk profiles and route components.

System Architecture / Workflow



Major Components

- ▶ Risk Scoring Engine
- ▶ Risk Classification Module
- ▶ Road Network Graph and Route

Data / Control Flow

- ▶ Environmental, road data collected.
- ▶ Risk calculation done.
- ▶ Risk zones and routes are displayed.

Technology Stack & Methodology

Technology Stack

- ▶ Programming: C++
- ▶ Concepts: OOP and DSA
- ▶ Data: CSV / JSON
- ▶ Web: HTML, CSS, JavaScript
- ▶ Map: Leaflet and OpenStreetMap
- ▶ Version Control: Git / GitHub

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

C++ implements OOP and DSA, while Graphs and Priority Queues support route processing, and Leaflet with OpenStreetMap provides map visualization.

Initial Progress & Team Contribution

Progress Achieved

- ▶ Background and problem study completed
- ▶ Project requirements identified
- ▶ Landslide risk factors identified
- ▶ System architecture prepared
- ▶ DSA-based route approach selected
- ▶ Technology stack identified
- ▶ Project proposal completed

Individual Contribution

Member	Role	Contribution
Smriti Singh	TeamLead / Developer	Frontend,documentation and project planning
Jahnvi Verma	Developer	DSA and route recommendation
Sanya Verma	Risk Analysis	Risk factors, data organization and system workflow
Anshika Gaur	Backend Developer	Backend andintegration,

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- ▶ Risk assessment and route recommendation system
- ▶ Risk scores and route results
- ▶ Safer route support
- ▶ Real-time data integration

References

1. OpenStreetMap contributors – OpenStreetMap.
2. Leaflet.js documentation – Leaflet.
3. Cormen, Leiserson, Rivest and Stein – Introduction to Algorithms.
4. Relevant landslide hazard and geological survey publications.

Thank You
Questions & Suggestions